

Minimization exercises, large implicants

Exercise 1

ab / cd	00	01	11	10
00	1	1	0	1
01	1	1	0	1
11	1	1	1	0
10	1	1	1	1

Exercise 2

ab / cd	00	01	11	10
00	1	1	1	0
01	1	0	0	0
11	1	1	0	1
10	1	1	1	1

Exercise 3

ab / cd	00	01	11	10
00	0	1	1	1
01	1	1	1	0
11	1	1	1	1
10	0	0	1	1

Exercise 4

ab / cd	00	01	11	10
00	0	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	0

Exercise 5

ab / cd	00	01	11	10
00	0	0	1	0
01	1	0	1	0
11	1	1	1	1
10	1	0	1	1

Exercise 6

ab / cd	00	01	11	10
00	1	0	1	1
01	1	1	1	1
11	1	1	1	0
10	1	1	0	1

Exercise 7

ab / cd	00	01	11	10
00	1	1	0	1
01	1	0	1	1
11	1	1	1	1
10	1	1	1	0

Exercise 8

ab / cd	00	01	11	10
00	0	0	1	0
01	0	0	1	1
11	0	1	1	1
10	0	0	1	1

Exercise 9

ab / cd	00	01	11	10
00	0	1	1	0
01	1	0	1	1
11	1	0	0	1
10	1	1	1	1

Exercise 10

ab / cd	00	01	11	10
00	1	1	0	0
01	0	0	0	0
11	1	1	1	1
10	0	1	0	1

Exercise 1

ab / cd	00	01	11	10
00	1	1	0	1
01	1	1	0	1
11	1	1	1	0
10	1	1	1	1

Solution to Exercise 1

There are 5 prime implicants:

$$\begin{aligned} &\bar{c} \\ &a\bar{b} \\ &ad \\ &\bar{a}\bar{d} \\ &\bar{b}\bar{d} \end{aligned}$$

A minimum cover is made of 4 prime implicants:

$$\bar{c} + ad + \bar{a}\bar{d} + \bar{b}\bar{d}$$

Exercise 2

ab / cd	00	01	11	10
00	1	1	1	0
01	1	0	0	0
11	1	1	0	1
10	1	1	1	1

Solution to Exercise 2

There are 6 prime implicants:

$$\begin{aligned} &a\bar{b} \\ &a\bar{c} \\ &\bar{b}\bar{c} \\ &\bar{b}d \\ &a\bar{d} \\ &\bar{c}\bar{d} \end{aligned}$$

A minimum cover is made of 4 prime implicants:

$$a\bar{c} + \bar{b}d + a\bar{d} + \bar{c}\bar{d}$$

Exercise 3

ab / cd	00	01	11	10
00	0	1	1	1
01	1	1	1	0
11	1	1	1	1
10	0	0	1	1

Solution to Exercise 3

There are 7 prime implicants:

$$a b$$

$$a c$$

$$\bar{b} c$$

$$b \bar{c}$$

$$\bar{a} d$$

$$b d$$

$$c d$$

A minimum cover is made of 4 prime implicants:

$$\bar{b} c + b \bar{c} + \bar{a} d + a c$$

Exercise 4

ab / cd	00	01	11	10
00	0	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	0

Solution to Exercise 4

There are 4 prime implicants:

$$b$$

$$d$$

$$\bar{a} c$$

$$a \bar{c}$$

A minimum cover is made of 4 prime implicants:

$$b + d + \bar{a} c + a \bar{c}$$

Exercise 5

ab / cd	00	01	11	10
00	0	0	1	0
01	1	0	1	0
11	1	1	1	1
10	1	0	1	1

Solution to Exercise 5

There are 5 prime implicants:

$$ab$$

$$ac$$

$$cd$$

$$a\bar{d}$$

$$b\bar{c}\bar{d}$$

A minimum cover is made of 4 prime implicants:

$$ab + cd + a\bar{d} + b\bar{c}\bar{d}$$

Exercise 6

ab / cd	00	01	11	10
00	1	0	1	1
01	1	1	1	1
11	1	1	1	0
10	1	1	0	1

Solution to Exercise 6

There are 8 prime implicants:

$$\bar{a}b$$

$$\bar{a}c$$

$$a\bar{c}$$

$$b\bar{c}$$

$$bd$$

$$\bar{a}\bar{d}$$

$$\bar{b}\bar{d}$$

$$\bar{c}\bar{d}$$

A minimum cover is made of 5 prime implicants:

$$\bar{a}c + a\bar{c} + bd + \bar{b}\bar{d} + \bar{c}\bar{d}$$

Exercise 7

ab / cd	00	01	11	10
00	1	1	0	1
01	1	0	1	1
11	1	1	1	1
10	1	1	1	0

Solution to Exercise 7

There are 8 prime implicants:

$$a b$$

$$b c$$

$$a \bar{c}$$

$$\bar{b} \bar{c}$$

$$a d$$

$$\bar{a} \bar{d}$$

$$b \bar{d}$$

$$\bar{c} \bar{d}$$

A minimum cover is made of 5 prime implicants:

$$b c + \bar{b} \bar{c} + a d + \bar{a} \bar{d} + \bar{c} \bar{d}$$

Exercise 8

ab / cd	00	01	11	10
00	0	0	1	0
01	0	0	1	1
11	0	1	1	1
10	0	0	1	1

Solution to Exercise 8

There are 4 prime implicants:

$$a c$$

$$b c$$

$$c d$$

$$a b d$$

A minimum cover is made of 4 prime implicants:

$$a c + b c + c d + a b d$$

Exercise 9

ab / cd	00	01	11	10
00	0	1	1	0
01	1	0	1	1
11	1	0	0	1
10	1	1	1	1

Solution to Exercise 9

There are 6 prime implicants:

$$a\bar{b}$$

$$\bar{b}d$$

$$a\bar{d}$$

$$b\bar{d}$$

$$\bar{a}bc$$

$$\bar{a}cd$$

A minimum cover is made of 4 prime implicants:

$$\bar{b}d + b\bar{d} + a\bar{d} + \bar{a}cd$$

Exercise 10

ab / cd	00	01	11	10
00	1	1	0	0
01	0	0	0	0
11	1	1	1	1
10	0	1	0	1

Solution to Exercise 10

There are 5 prime implicants:

$$ab$$

$$\bar{a}\bar{b}\bar{c}$$

$$a\bar{c}d$$

$$\bar{b}\bar{c}d$$

$$ac\bar{d}$$

A minimum cover is made of 4 prime implicants:

$$ab + \bar{a}\bar{b}\bar{c} + ac\bar{d} + \bar{b}\bar{c}d$$